WEATHER

Jupiter has no notable seasons, and the planet's temperature is virtually uniform. Its polar regions have temperatures similar to those of its equatorial regions because of internal heating. Jupiter radiates about 1.7 times more heat than it absorbs from the Sun. The excess is infrared heat left from when the planet was formed. Most of Jupiter's weather occurs in the part of its atmosphere that contains its distinct white and red-brown cloud layers and is dominated by clouds, winds, and storms. The rising warm air and descending cool air within the atmosphere produce winds that are channeled around the planet, both to the east and west, by Jupiter's fast spin. The wind speed changes with latitude; winds within the equatorial region are particularly strong and reach speeds in excess of 250 mph (400 kph). The solar and infrared heat, the wind, and Jupiter's spin combine to produce regions of turbulent motion, including circular and oval cloud structures, which are giant storms. The smallest of these storms are like the largest hurricanes on Earth. They can be relatively short-lived and last for just days at a time, but others endure for years. Jupiter's most prominent feature, the Great Red Spot, is an enormous high-pressure storm that may have first been sighted from Earth over 340 years ago.

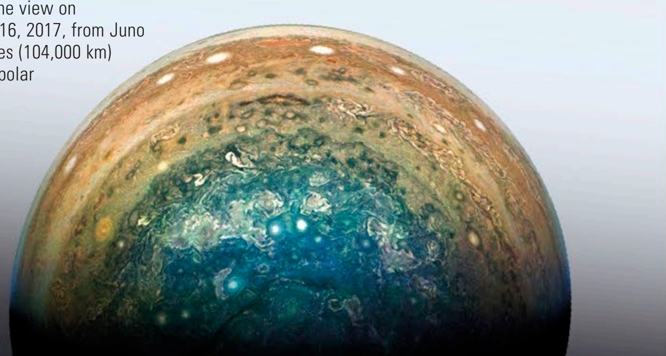
THE GREAT RED SPOT
This giant storm, which is bigger than Earth, is constantly changing its size, shape, and color. It rotates counterclockwise every 6 to 7 days.

EXPLORING SPACE

THE POLES

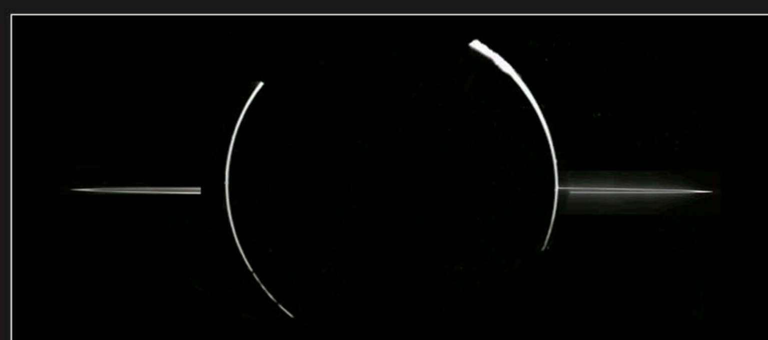
Jupiter's poles had only ever been seen at an angle until NASA's Juno probe arrived in polar orbit in 2016. The familiar belt and zone pattern is replaced by spiral storms, known as cyclones. In 2016, the south polar cyclone was surrounded by a polygonal pattern of five circumpolar cyclones (of which three are in sunlight in the accompanying image), whereas the north pole had a pattern of eight circumpolar cyclones. By late 2019, a sixth southern circumpolar cyclone had appeared.

JUPITER'S SOUTH POLE
This was the view on December 16, 2017, from Juno 65,000 miles (104,000 km) above the polar cloudtops.



RINGS

Jupiter's ring system was revealed for the first time in an image taken by Voyager 1 in 1979. It is a thin, faint system made of dust-sized particles knocked off Jupiter's four inner moons. The system consists of three parts. The main ring is flat and is about 4,350 miles (7,000 km) wide and less than 18 miles (30 km) thick. Outside this is the flat gossamer ring, which is 528,000 miles (850,000 km) wide and stretches beyond Amalthea to Thebe's orbit. On the inside edge of the main ring is the 12,400-mile (20,000-km) thick doughnut-shaped halo. Its tiny dust grains reach down to Jupiter's cloudtops.



JUPITER'S MAIN RING
Jupiter's main ring was imaged by Galileo with the Sun behind the planet. From this position, small particles within the ring and in Jupiter's upper atmosphere stand out. The halo and gossamer ring are revealed only if the main ring is overexposed.